Lecture plan · for the lecturer

The relational model and normalisation

Data engineering · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Use keys and dependencies.
- Normalise to 3NF and BCNF.
- Teach from anomalies.

Activity: Break it, then fix it

1. Warm-up (5 min): Show the enrolment table with a lecturer's two phone numbers and ask which is right.
2. Break (15 min): Groups insert, update and delete in the flat table and list what fails.
3. Derive (15 min): State the dependencies and compute the key by closure.
4. Fix (20 min): Decompose to 3NF and check the join rebuilds the original.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which anomaly appeared first?
- Where did the dependencies come from?
- When would you denormalise?

Exit ticket

- Write one functional dependency from a system you use.

Try it in your next lecture
Give groups a flat enrolment spreadsheet; each performs one insert, one update and one delete and reports what went wrong, then derive the dependencies together.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Course → lecturer → office, keyed by course. Is it in 3NF?**

◯ Only if office is unique ◯ Yes ◯ No, the dependency is transitive

2 **Compute a closure.**

With A → B and B → C, the closure of {A} is {A, B, _____}, so A is a _____ of R(A, B, C).

Word bank: dependency · C · key · D

3 **Enrol(student, course, grade, lecturer, phone) with course → lecturer → phone and key (student, course). Which are true? Choose all that apply.**

☐ It violates 3NF ☐ It violates 2NF ☐ It is in BCNF

☐ Phone changes cause update anomalies

4 **Some relations cannot be decomposed into BCNF while preserving all functional dependencies.**

◯ True ◯ False

5 **Match each normal form to what it forbids.**

1NF	• • • •	Transitive dependence on the key
2NF	• • • •	Any determinant that is not a superkey
3NF	• • • •	Repeating groups and non-atomic values
BCNF	• • • •	Dependence on part of a key

Answers · for the lecturer

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1 **No, the dependency is transitive**

Office depends on a non-key attribute.

2 **With A → B and B → C, the closure of {A} is {A, B, C}, so A is a key of R(A, B, C).**

A determines B, and B determines C, so A determines every attribute.

3 **It violates 2NF; It violates 3NF; Phone changes cause update anomalies**

Lecturer depends on part of the key (course), breaking 2NF and hence 3NF and BCNF.

4 **True**

Lossless BCNF is always possible, but dependency preservation is not; 3NF guarantees both.

5 **1NF — Repeating groups and non-atomic values**
2NF — Dependence on part of a key
3NF — Transitive dependence on the key
BCNF — Any determinant that is not a superkey

Each form tightens the one before.

For you to think about

Do your students meet anomalies before they meet normal-form definitions? What table could you break in class?